

Explainable AI

Banana Assignment - 3

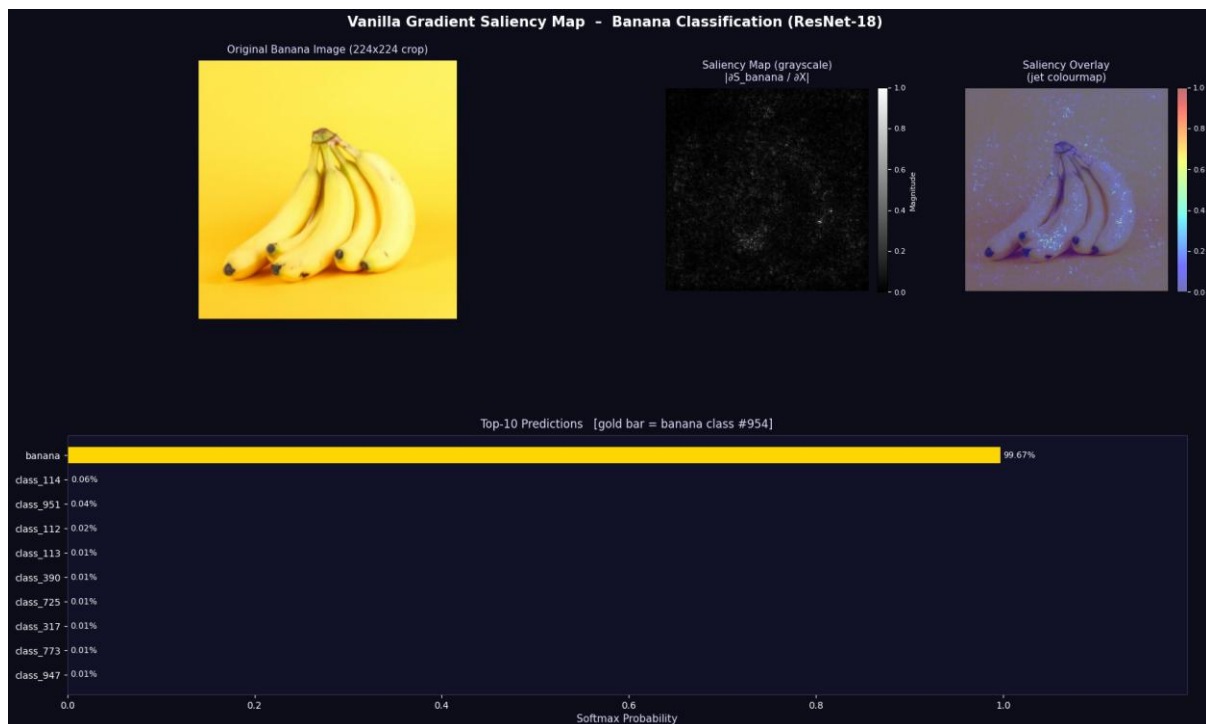
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Task 1: Vanilla Gradient Saliency Map for Banana Classification

A pretrained ResNet-18 model (ImageNet) was used to compute a Vanilla Gradient Saliency Map for the banana class (index 954). The input banana image was preprocessed and passed through the model. Gradients of the banana class score with respect to the input pixels were computed using a single backward pass. The absolute gradient values were taken and collapsed across RGB channels using max-pooling, then normalized to [0,1] to produce the saliency map.

The model classified the banana image with **99.67% confidence**. The saliency map revealed that **edges and curved tips of the banana were most highlighted**, confirming that the model focuses on boundary regions and class-discriminative features. The flat interior showed moderate activation while the background remained near-zero.

Output :



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Task 2: Smooth Visualization via Activation Maximization

Starting from a random noise image, gradient ascent was applied to optimize the input pixels to maximally activate the banana class neuron in ResNet-18. At each iteration, the banana class score was computed via a forward pass and gradients were backpropagated to update the image — not the model weights. To ensure smooth visualization, Gaussian blur was applied every 8 steps and L2 regularization was used to prevent pixel explosion.

After 512 iterations, the banana class score rose from **1.05 to 124.99**, with a final probability of **99.45%**. The generated image exhibited dominant yellow/warm tones and curved edge-like structures, confirming that ResNet-18 internally associates the banana class with warm colours, curved geometry, and textured surfaces.

Output :

